

# David Cornelis Groothuizen Dijkema

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## Education

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### University of Auckland | Waipapa Taumata Rau

Doctoral Candidate in Mathematics Mar. 2022 — Aug. 2025

Thesis: Analysis of Dynamics near Heteroclinic Networks using Projected Maps

Supervisors: Profs. Claire Postlethwaite and Vivien Kirk

*Thesis submitted August 26<sup>th</sup>, 2025, currently under examination*

Bachelor of Science (Honours) 2021

*with First Class Honours in Applied Mathematics*

Honours dissertation: A heteroclinic network in a spatially-extended four species interaction model

Supervisor: Prof. Claire Postlethwaite

Bachelor of Engineering (Honours) 2017 — 2020

*with First Class Honours in Engineering Science*

Honours dissertation: Synthetic-aperture radar processing on the NVidia Jetson AGX Xavier

Supervisors: Prof. John Cater and Assoc Prof. Nicholas Rattenbury

## Employment history

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### University of Auckland | Waipapa Taumata Rau

Professional Teaching Fellow

Department of Mathematics

Sep. 2025 —

## Publications and preprints

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David C Groothuizen Dijkema and Claire M Postlethwaite (2023). Travelling waves and heteroclinic networks in models of spatially-extended cycle competition. *Nonlinearity*, **36**(12):6546–6588.

DOI: [10.1088/1361-6544/ad0212](https://doi.org/10.1088/1361-6544/ad0212). arXiv: [2208.05630](https://arxiv.org/abs/2208.05630) [[math.DS](#)].

David C Groothuizen Dijkema, Vivien Kirk, and Claire M Postlethwaite. Analysis of dynamics near heteroclinic networks in  $\mathbb{R}^4$  with a projected map. *Submitted to SIAM Journal on Applied Dynamical Systems*.

arXiv: [2410.21486](https://arxiv.org/abs/2410.21486) [[math.DS](#)].

David C Groothuizen Dijkema, Claire M Postlethwaite, and Alastair M Rucklidge. Continuity of projected maps near heteroclinic networks in  $\mathbb{R}^4$ . *In preparation. Preprint available upon request.*

David C Groothuizen Dijkema, Claire M Postlethwaite, and David J W Simpson. Strings of sausages and regular cycling near the Rock-Paper-Scissors-Lizard-Spock network. *In preparation.*

## Awards and scholarships

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Honourable mention for the Australia and New Zealand Industrial and Applied Mathematics T M Cherry Prize for Best Student Talk at the ANZIAM Conference | 2025             |
| New Zealand Mathematical Society Aitken Prize for Best NZMS Student Talk at the Joint Meeting of the NZMS, AustMS, and AMS                              | 2024             |
| New Zealand Mathematics Research Institute Kalman Summer Scholarship<br>(To support attendance at the NZMRI 2024 Summer School)                         |                  |
| Nonlinearity Outstanding Reviewer Award                                                                                                                 | 2023             |
| University of Auckland Academic Career Advancement Award<br>(To participate in the Academic Career Advancement Programme)                               |                  |
| Australia and New Zealand Industrial and Applied Mathematics Prize<br>for Best Poster Presentation at the NZMS Colloquium                               |                  |
| Honourable mention for the New Zealand Mathematical Society Aitken Prize<br>for Best Student Talk at the NZMS Colloquium                                |                  |
| Royal Society of New Zealand Te Apārangi Marsden Fund Doctoral Scholarship<br>(Full scholarship, including stipend and fees)                            | 2022             |
| University of Auckland Department of Mathematics Student Research Conference prize<br>First in Course Award<br>for MATHS 761 (Dynamical Systems)        | 2021             |
| University of Auckland Postgraduate Honours Scholarship<br>(Full scholarship, including stipend and fees)                                               |                  |
| University of Auckland Summer Research Scholarship                                                                                                      | 2020             |
| First in Course Award<br>for ENGSCI 355 (Applied Modelling in Simulation and Optimisation)                                                              | 2019             |
| University of Auckland Faculty of Engineering Dean's Honours List                                                                                       | 2017, 2018, 2019 |

## Teaching experience

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| <b>Department of Mathematics, Faculty of Science, University of Auckland</b>                                                                                                                                                                       | 2022 — 2024 |
| Teaching support                                                                                                                                                                                                                                   |             |
| Developed resources to support the use of the software <a href="#">CoCo</a> in MATHS 761 (Dynamical Systems), a postgraduate course. Developed resources include MATLAB templates and completed examples for student use, and tutorial worksheets. |             |
| Graduate Teaching Assistant for                                                                                                                                                                                                                    |             |
| COMPSCI 120 (Mathematics for Computer Science)                                                                                                                                                                                                     |             |
| COMPSCI 225 (Discrete Structures in Mathematics and Computer Science)                                                                                                                                                                              |             |
| MATHS 108 (General Mathematics 1)                                                                                                                                                                                                                  |             |
| MATHS 120 (Algebra 1)                                                                                                                                                                                                                              |             |
| MATHS 130 (Calculus 1)                                                                                                                                                                                                                             |             |
| MATHS 253 (Algebra and Calculus 3)                                                                                                                                                                                                                 |             |
| MATHS 260 (Differential Equations)                                                                                                                                                                                                                 |             |
| Mathematics Assistance Room (an undergraduate drop-in clinic)                                                                                                                                                                                      |             |
| <b>Department of Engineering Science and Biomedical Engineering,<br/>Faculty of Engineering, University of Auckland</b>                                                                                                                            | 2020        |
| Teaching Assistant for                                                                                                                                                                                                                             |             |
| ENGSCI 255 (Modelling and Analytics in Operations Research)                                                                                                                                                                                        |             |
| ENGSCI 211 (Mathematical Modelling 2)                                                                                                                                                                                                              |             |

## Service

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| Co-organiser of a minisymposium at SIAM DS 25                                                                                                                                                                                                                                                                                          | 2025           |
| I co-organised a two-part minisymposium titled <i>Dynamics near Homoclinic and Structures</i> with Alexander Lohse, Universität Hamburg, Germany at the SIAM Conference on Applications of Dynamical Systems 2025, Denver, CO, USA                                                                                                     |                |
| Chair of the Social Committee of the Joint Meeting of the NZMS, AustMS, and AMS                                                                                                                                                                                                                                                        | 2024           |
| I served as the chair of a committee composed of PhD students and postdocs of the Mathematics Department to support the conference organising committee. We organised a mentoring programme, a social event for students and postdocs, a daily newsletter, and a WhatsApp community, and compiled recommendations of local activities. |                |
| Co-organiser of the Antipodeal Dynamics Workshop                                                                                                                                                                                                                                                                                       | 2023           |
| I co-organised the virtual workshop with Prof. Claire Postlethwaite, and Prof. Peter Ashwin and Muhammed Fadera (both University of Exeter, UK), which featured invited presentations by Prof. Krasimira Tsaneva-Atanasova, Assoc. Prof. David Simpson, and Assist. Prof. Alexandre Rodrigues, and a virtual poster session.           |                |
| Reviewer for Nonlinearity and Journal of Nonlinear Science                                                                                                                                                                                                                                                                             | 2023 – present |

## Professional development

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| <b>Academic Career Advancement Programme (ACAP)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2024 |
| A year-long course run by the School of Graduate Studies at the University of Auckland for 20 doctoral candidates, featuring 13 three-hour sessions covering different aspects of an academic career. Ten of these sessions were organised by ACAP participants in pairs. I co-organised a seminar on the <i>Broader Dissemination of Research</i> , which included a one-hour workshop with Prof. Neal Curtis on using ideas from comics to design posters, and a talk by Prof. Cliona Ní Nhurchú on engaging with communities, industry, and government. |      |
| <b>Department of Mathematics Tutor Training</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2022 |
| A three-hour workshop run by the Department, focussing on the fundamentals of interacting with students in tutorials, being an engaging and effective tutor, and cultural sensitivity.                                                                                                                                                                                                                                                                                                                                                                     |      |

## Research visits

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| <b>School of Mathematics, University of Leeds, UK</b>                                                                                         | Jun. 5 – Jul. 7, 2023 |
| Visited Prof. Alastair Rucklidge to work on understanding the continuity of projected maps of heteroclinic networks in $\mathbb{R}^4$ .       |                       |
| <b>Centre of Mathematics, University of Porto, Portugal</b>                                                                                   | Jun. 19 – 23, 2023    |
| Visited Prof. Sofia Castro to work on various problems concerning the dynamics of trajectories near heteroclinic networks in $\mathbb{R}^4$ . |                       |

## Presentations

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| Analysing dynamics near heteroclinic networks with a projected map.<br><i>XLV Dynamics Days Europe</i> . Thessaloniki, Greece                                                                                               | Jun. 2025 |
| Analysing dynamics near heteroclinic networks with a projected map.<br><i>Minisymposium on Dynamics Near Homoclinic and Heteroclinic Structures, SIAM Conference on Applications of Dynamical Systems</i> . Denver, CO, USA | May 2025  |

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| Analysing dynamics near heteroclinic networks with a projected map.<br><i>Australia and New Zealand Industrial and Applied Mathematics Conference.</i> Coffs Harbour, NSW, Australia.                                                                       | Feb. 2025 |
| Analysing dynamics near heteroclinic networks with a projected map.<br><i>Special session on Ergodic Theory and Dynamical Systems, Joint Meeting of the NZMS, AustMS, and AMS.</i> University of Auckland, NZ.                                              | Dec. 2024 |
| Travelling waves in spatially-extended models of cyclic competition.<br><i>Pattern Formation Workshop.</i> University of Auckland, NZ.                                                                                                                      | Dec. 2024 |
| Dynamics near heteroclinic networks.<br><i>Applied Mathematics Unit Doctoral Talks.</i> University of Auckland, NZ.                                                                                                                                         | Oct. 2024 |
| Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system.<br><i>Australia and New Zealand Industrial and Applied Mathematics Conference.</i> Hahndorf, SA, Australia.                                                        | Feb. 2024 |
| Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system.<br><i>New Zealand Mathematical Society Colloquium.</i> Victoria University of Wellington, NZ.                                                                      | Dec. 2023 |
| Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition.<br><i>New Zealand Mathematics and Statistics Postgraduate Conference.</i> Wellington, NZ.                                                                    | Nov. 2023 |
| Dynamics near heteroclinic cycles and networks.<br><i>Leeds Applied Nonlinear Dynamics Seminar.</i> University of Leeds, UK.                                                                                                                                | Jun. 2023 |
| Dynamics near heteroclinic cycles and networks.<br><i>Centro de Matemática da Universidade do Porto Seminar.</i> University of Porto, Portugal.                                                                                                             | Jun. 2023 |
| Switching near heteroclinic networks as a piecewise-smooth dynamical system.<br><i>Minisymposium on Long-Term Dynamics: Attraction and Stability Properties of Invariant Sets, SIAM Conference on Applications of Dynamical Systems.</i> Portland, OR, USA. | May 2023  |
| Switching near heteroclinic networks as a piecewise-smooth system.<br><i>Sydney Dynamics Group Workshop: Dynamics in NZ.</i> Karioitahi, NZ.                                                                                                                | Nov. 2022 |
| Switching near heteroclinic networks as a piecewise-smooth system.<br><i>Applied Mathematics Unit Doctoral Talks.</i> University of Auckland, NZ.                                                                                                           | Oct. 2022 |
| Stability and cycling near heteroclinic cycles and networks.<br><i>Department of Mathematics Student Research Conference.</i> University of Auckland, NZ.                                                                                                   | Jun. 2022 |
| Stability and cycling near heteroclinic cycles and networks.<br><i>Department of Mathematics Postgraduate Seminar.</i> University of Auckland, NZ.                                                                                                          | Jun. 2022 |
| Spatially-extended models of competition between four species.<br><i>Department of Mathematics Postgraduate Mid-year Talks.</i> University of Auckland, NZ.                                                                                                 | Jul. 2021 |
| A spatially-extended model of competition between four species.<br><i>Department of Mathematics Student Research Conference.</i> University of Auckland, NZ.                                                                                                | Jun. 2021 |
| Heteroclinic networks in partial differential equations.<br><i>Department of Mathematics Summer Research Conference.</i> University of Auckland, NZ.                                                                                                        | Feb. 2021 |

## Posters

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| Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition.<br><i>New Zealand Mathematical Society Colloquium.</i> Victoria University of Wellington, NZ. | Dec. 2023 |
| Switching near heteroclinic networks as a piecewise-smooth dynamical system.<br><i>Antipodeal Dynamics Workshop.</i> University of Exeter, UK and University of Auckland, NZ.                | Oct. 2022 |

## Conferences and workshops attended

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| Gene Golub SIAM Summer School, on <i>frontiers in multi-dimensional pattern formation</i>          | Aug. 2025 |
| XLV Dynamics Days Europe                                                                           | Jun. 2025 |
| SIAM Conference on Applications of Dynamical Systems                                               | May 2025  |
| Australia and New Zealand Industrial and Applied Mathematics Conference                            | Feb. 2025 |
| Joint Meeting of the NZMS, AMS, and AustMS                                                         | Dec. 2024 |
| Pattern Formation Workshop                                                                         | Dec. 2024 |
| (Satellite meeting to the Joint Meeting of the NZMS, AMS, and AustMS.)                             |           |
| Australia and New Zealand Industrial and Applied Mathematics Conference                            | Feb. 2024 |
| New Zealand Mathematics Research Institute Summer School, on <i>mathematical neuroscience</i>      | Jan. 2024 |
| (Received the Kalman Summer Scholarship to support attendance.)                                    |           |
| Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)            | Dec. 2023 |
| New Zealand Mathematical Society Colloquium                                                        | Dec. 2023 |
| New Zealand Mathematics and Statistics Postgraduate Student Conference                             | Nov. 2023 |
| SIAM Conference on Applications of Dynamical Systems                                               | May 2023  |
| New Zealand Mathematics Research Institute Summer School, on <i>groups, geometry, and dynamics</i> | Jan. 2023 |
| Sydney Dynamics Group Workshop: Dynamics in NZ                                                     | Nov. 2022 |
| Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)            | Oct. 2022 |
| Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)                  | Feb. 2022 |
| Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)            | Oct. 2021 |
| SIAM Conference on Applications of Dynamical Systems (Virtual)                                     | May 2021  |
| Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)                  | Feb. 2021 |

## Professional affiliations

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**New Zealand Mathematical Society**  
**Australia New Zealand Industrial and Applied Mathematics**  
**Society for Industrial and Applied Mathematics**  
Student member

## Other employment history

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|-------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Department of Mathematics, Faculty of Science, University of Auckland</b>                                                  |                       |
| Graduate Teaching Assistant                                                                                                   | Jul. 2022 — Jun. 2024 |
| Research Assistant                                                                                                            | Nov. 2021 — Jun. 2022 |
| 38 Princes St, Auckland Central, Auckland, 1010, NZ                                                                           |                       |
| <b>Department of Engineering Science and Biomedical Engineering,</b><br><b>Faculty of Engineering, University of Auckland</b> |                       |
| Teaching Assistant                                                                                                            | Mar. 2020 — Nov. 2020 |
| 70 Symonds Street, Grafton, Auckland, 1010, NZ                                                                                |                       |
| <b>SUEZ Smart Solutions Ltd.</b>                                                                                              | Nov. 2019 — Mar. 2020 |
| Assistant Optimisation Engineer                                                                                               |                       |
| AMI House, 63 Albert St, Auckland Central, Auckland, 1010, NZ                                                                 |                       |
| <b>Watercare Services Ltd.</b>                                                                                                | Oct. 2018 — Nov. 2019 |
| Undergraduate Engineer                                                                                                        |                       |
| 73 Remuera Road, Remuera, Auckland, 1050, NZ                                                                                  |                       |